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NPS/001/009 – Technical Specification for 11kV, 20kV and 33kV Pole Mounted Auto-reclose Circuit Breakers.

1. Purpose

The purpose of this document is to detail the requirements for 11kV, 20kV and 33kV pole mounted auto-reclose circuit breakers for use on the overhead line distribution networks of NEDL and YEDL.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
NE AR1	2003	Specification for 11kV, 20kV and 33kV Pole Mounted Auto-Reclose Circuit Breakers

2. Scope

This document applies to the design and constructional requirements for 11kV, 20kV and 33kV pole mounted auto-reclosers and associated equipment.

The equipment shall be in compliance with the national and international standards detailed in this document.

3. Technical Requirements

3.1 Ratings

3.1.1 Rated Voltage (Ur)

The equipment shall be rated for use at 12k, 24kV & 36kV as defined in the Range 1, Series 1 sub clause 4.1.1 of IEC 60694.

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3.1.2 Rated Insulation Level (Up)

Rated Voltage Ur kV(rms. value)	Rated short-duration power frequency withstand voltage Ud kV (rms. value)				Rated lightning impulse withstand voltage Up kV (peak)	
	Common Value		Across the isolating distance		Common Value	Across the isolating distance
	Dry	Wet	Dry	Wet		
12	50	45	55	50	110	125
24	70	60	77	66	125 (**150)	145 (**165)
36	95	80	105	88	200	220

** The preferred impulse insulation withstand voltage shall be as detailed in Table 1b of IEC 60694, however the reduced value detailed in Table 1.1a of ENATS 41-36 will be acceptable for 20kV units.

3.1.3 Current Ratings

Voltage	Normal Current	Rated short-time withstand current (kA)
12kV	630 A	12.5kA for 3 secs
24kV		
36kV	800 A	16 kA for 3 secs

3.1.4 Values of Rated Line-charging Breaking Currents and Rated Cable-charging Breaking Currents

Rated Voltage kV	Breaking Current KA	
	Rated Cable Charging	Rated Line Charging
12kV	25	10
24kV	31.5	
36kV	50	

3.1.5 Rated Frequency

The rated frequency shall be 50Hz.

3.1.6 Electromagnetic Compatibility

Equipment (including control and communication systems) shall comply with EMC Directive 89/336/EEC, IEC60694 and BS EN 62271.

3.1.7 Arc Interruption Medium

Where switch arcs are not broken in air, the arc interruption medium shall be non-oil.

In the case of SF₆ filled equipment the maximum and minimum permissible gas pressure values shall be permanently marked on the gauge for example as absolute pressure values in P.S.I. or N/m² or as red and green zones on a Bourdon tube gauge. Low-pressure lockout facilities shall be provided, unless the SF₆ gas is used solely for insulation purposes and the equipment has been tested for satisfactory operation with the gas at atmospheric pressure.

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3.1.8 System Earthing

The equipment is required for use on a three phase, 50hz, 11, 20 or 33kV electrical distribution network that has its neutral directly earthed, resistance earthed or earthed via an arc suppression coil.

3.1.9 Degree of Protection by Enclosures

Protection Against Access - The degree of protection (IEC 60529) of persons provided by the enclosure against access to hazardous parts and protection of the equipment against ingress of solid foreign objects shall not be less than IP3XD.

Protection Against Ingress of Moisture - The degree of protection for pole-mounted enclosed equipment shall be not less than IP34. The weatherproofing test method shall be the method given in annex C of IEC 60694.

3.2 General Requirements

3.2.1 Design of Control Cubical

The control cubicle and doors shall be fabricated from stainless steel. The cubicles shall be designed to meet the "degrees of protection" requirements detailed in clause 3.1.9 of this specification.

The control cubicle shall incorporate a 12mm stud mounted externally in the base to allow a minimum of 2 earth connections to be made to the cubicle.

Additionally the base of the cubicle shall allow external power supplies and communications aerials to be terminated into the base of the cubicle. Specifically it shall incorporate a M20 hole for the auxiliary power supply cable and an M16 hole for the communications aerial.

The control cubicle shall be designed to be mounted on a wood pole and be supplied with all brackets and fixings required.

The control cubicle shall be large enough to incorporate the following equipment within the cubicle:

- Control Relay
- Communications RTU,
- Battery charger and batteries,
- Heater and thermostatic control unit,
- Radio communications tray to allow a radio or modem unit to be installed.

3.2.2 Cubical and Circuit Breaker Connection

The circuit breaker shall be connected to the control cubicle by an umbilical cable. The cable shall be provided in a standard 7m length with an option for an 11m version.

The cable shall connect the recloser to the cubicle and be designed so as to short out the recloser CT's when the cable is unplugged.

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3.2.3 Operational Safety and Locking

The control cabinet and relay unit shall be designed to allow a field operative to place the equipment into an open or closed status. Once in the designated status, it shall not be possible to change the status of the equipment by remote control means. The control cubicle door(s) shall be suitable for application of a padlock with a 7mm diameter shackle.

3.2.4 Power Source Requirements

Control cubicles shall include battery charger units and secondary batteries. The batteries shall be low loss rechargeable 12V batteries that have a minimum operational lifetime of 10 years, in the designed environment.

The capacity of the secondary batteries shall be such that the equipment remains fully functional for at least two weeks, even if the charging devices should fail. The test duty cycle for this two week period shall be:

- Quiescent for 366 hours, followed by OPEN - 3s - CLOSE - 3s - OPEN - 3s - CLOSE.

Typically the Charging units will be supplied from externally mounted auxiliary VT's (system voltage /230V), although they may also be supplied from local 400/230V 50Hz supply networks. All charging units shall incorporate battery condition and test software.

In the event of a failure of the external supply voltage or of the battery charger or battery, a means shall be provided to generate an alarm that registers both locally and to the RTU located in the control cubicle for remote indication.

Control cabinets shall incorporate double pole isolators in the LV power supply circuit. The cabinet shall have the facility to isolate potential LV supplies from a single point.

Disconnection of batteries for maintenance or replacement purposes, or in the event of battery/charger/supply failure causing the batteries to discharge, shall not cause a loss of stored data within the Relay or RTU unit.

3.2.5 Stored Energy

The circuit breaker shall not be allowed to close unless there is sufficient stored charge available to allow the circuit breaker to be electrically tripped.

3.2.6 Heaters

The control cabinets shall be supplied complete with thermostatically controlled heaters designed to work solely from an external 240V power supply. In the event of loss of external power supplies, they shall not impact on the standby capacity of the batteries.

3.2.7 Circuit Breaker Operational and Control Requirements

The following control functions shall be provided both locally and remotely through the SCADA system. At the local control panel the control functions will be provided by separate open and close switches, and separate protection control switches for each function or a multi position control switch.

The unit shall provide a separate toggle type remote/local switch and a separate live line switch.

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The following functions shall be provided for remote control use: -

- a) Close
- b) Open
- c) Protection - In/Out
- d) Sensitive Earth Fault (SEF) - In/Out
- e) Auto-reclose ON – Normal Automatic Sequence
- f) Auto-reclose OFF (Live Line) – Instantaneous Trip and Lockout
- g) Auto-reclose and instantaneous protection OFF (NON-AUTO) – 1 Delayed Trip and Lock out

The analogous above are to the selection facilities provided for in ACE report 54 (1977) – Report on Auto-reclosing Schemes for 11kV Distribution Networks.

3.2.8 Remote Operation of the Control Relay

The control relay must incorporate an integral RTU (Remote Terminal Unit). The RTU shall provide via a suitable communications system, a remote control and indication facility.

Additionally the RTU shall provide:-

- Support DNP3 level 2, 2002 - The DNP3 Template currently being utilised is shown in the table below. A replicated set of controls and indications shall be provided.
- Analogues that are 16 bit non assigned.
- Utilise a single bit for all controls and indications
- Data link confirms should only be initiated when transmitting multiple fragment messages.
- Radio power management - see clause 3.2.9

Function	Control	Indication	Alarm	Analogue	Notes
Open/ Close	Yes	Yes			Select/Execute cycle (SELEX)
Enable Protection/ Disable Protection	Yes		Yes*		Select/Execute cycle (SELEX)
Enable SEF Protection/ Disable SEF Protection	Yes		Yes*		Select/Execute cycle (SELEX)
Auto-Reclose On/ Auto-Reclose Off	Yes		Yes*		Select/Execute cycle (SELEX)
Reset Fault Flags and Currents	Yes		Yes*		Select/Execute cycle (SELEX)
Dummy CB Open/ Close	Yes		Yes*		Select/Execute cycle (SELEX)
OC Protection Operated			Yes		
EF Protection Operated			Yes		
SEF Operated			Yes		
High Current Alarm			Yes		
Local operation			Yes		
Remote Operation			Yes		

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Function	Control	Indication	Alarm	Analogue	Notes
230V Supply Fail			Yes		Requires a user settable time delay of not less than 180 Min in 1 min steps.
Battery Condition Poor			Yes		
Gas Pressure Low			Yes		
L1 Current (Amps)				Yes	Shall be capable of alarming for under/over current to limits set by an operator at the Control System
L2 Current (Amps)				Yes	Shall be capable of alarming for under/over current to limits set by an operator at the Control System
L3 Current (Amps)				Yes	Shall be capable of alarming for under/over current to limits set by an operator at the Control System
L1 Voltage (V)				Yes	Shall be capable of alarming for under/over voltage to limits set by an operator at the Control System
L2 Voltage (V)				Yes	Shall be capable of alarming for under/over voltage to limits set by an operator at the Control System
L3 Voltage (V)				Yes	Shall be capable of alarming for under/over voltage to limits set by an operator at the Control System

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Function	Control	Indication	Alarm	Analogue	Notes
230 V ac Voltage measurement (Volts)				Yes	
Operations Counter				Yes	
Clock					Clock shall be synchronised with the control system clock

* The indication of this control is a single bit indication and is therefore to be entered in to the template as an alarm.

3.2.9 Radio Power Management

The RTU unit shall incorporate a radio power management system. If the auxiliary power supply fails, the unit shall monitor the time of the failure and after a user settable time, disable the power supply to the radio. When the auxiliary power supply becomes available again, the RTU shall re-enable the power supply to the radio.

3.2.10 Computer Port

The electronic controller shall be supplied with a 25 way female D-type connector providing RS232 communication to the electronic controller.

3.2.11 Programming Software

Software shall be provided to control or programme the units via the computer port in clause 3.2.10 or via radio.

3.2.12 Protection Relay Characteristics

The relay shall meet the functional and environmental requirements detailed in ENA TS 48-5 "Electrical Relays" and any additional tests which may be deemed necessary due to the closing mechanism and the proximity of the recloser.

The relay shall provide the following protection functions:

- a) 2 pole Instantaneous Overcurrent (OC) and single pole Instantaneous Earth Fault.
- b) 2 pole Delayed Overcurrent (OC) and single pole Delayed Earth Fault with the following choice of characteristics:
 - i) Standard Inverse
 - ii) Very Inverse
 - iii) Extremely Inverse
 - iv) American National Standard Institute (ANSI)
 - v) Definite Time Lag
- c) Definite Time Lag Sensitive Earth Fault (SEF) - selected separately from b (v) above.

The protection shall remain stable under magnetising inrush and transient harmonic situations.

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The protection relay and RTU shall not lose or suffer corruption to any settings; in protection, control logic or communications in the event of a loss of power to the relay.

3.2.13 Protection – Functional Requirements

The recloser shall be equipped as follows:

- a) 4 trips to lockout with any combination of tripping characteristics.
- b) OC/EF and SEF sequences to be independent of each other.
- c) Selectable Sequence Co-ordination.
- d) Individually adjustable Dead Times
- e) Adjustable reclaim times
- f) Line check function
- g) Sequence co-ordination.
- h) Cold Load Pickup
- i) Broken conductor protection
- j) Directional Blocking
- k) Voltage Measurement on the incoming terminals with provision for an external CT for measurement of voltage on the load terminals.
- l) Power Measurement

Additionally the control unit shall allow the circuit breaker to operate like a sectionaliser. It shall provide accurate fault passage information via it's CT's to the RTU whilst remaining in it's closed state. It shall only change status as a result of a pre-programmed set of events or via remote operation.

3.2.14 Range of Protection Settings

Current Setting Ranges: -

Overcurrent – 50 to 600 Amps

Earth Fault – 10 to 200 Amps

Sensitive Earth Fault – 4 to 30 Amps

Instantaneous OC and EF with settings as the Overcurrent and Earth Fault Current detailed above.

Timing Range Settings: -

Overcurrent Time Multiplier - 0.1-1.0 TM

Overcurrent and Earth Fault Time Delay - 0.5 to 3.0 seconds

Instantaneous Time Delay - 0 to 2.0 seconds

Sensitive Earth Fault Time Delay - 0 to 20 seconds

Dead Time - 1 to 60 seconds

Reclaim Time - 5 to 60 seconds

Sequence co-ordination time - 0 to 1 seconds

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3.2.15 Indications Required Locally and Remote

Local Indication: -

- Number of recloser operations completed since installation.
- Fault current magnitudes and duration, for each phase, for at least the last 25 recloser operations
- Present load current.
- Maximum demand within the past 12 months.
- Contact wear on all three phases.
- Battery condition.

Remote Indication: -

- Indication of status, OPEN or CLOSED.
- Indication of current (three digits for current and five digits for voltage).
- Operation OPEN or CLOSE.
- Protection settings.
- Battery condition.
- Select individual protection feature e.g. SEF.

3.2.16 Input/Output Expander Card

Provision shall be made for an optional INPUT/OUTPUT expander card to provide optically isolated input contacts and voltage free output contacts. This may be required with externally provided RTU units or as a conversion medium between non-compliant DMP3 (detailed in Section 3.2.8 Table) communication equipment. Data maps and contact diagrams shall be available where this option is provided.

3.3 Recloser Requirements

3.3.1 Mounting Arrangements

The equipment shall be designed to allow mounting on single wood pole preferably utilising M20 pole bolts vertically spaced with 360mm centres. The preferred mounting arrangement will result in the recloser being located vertically below the overhead conductors.

3.3.2 HV Connections

On the 11 & 20kV units 400A mechanical parallel groove type connectors shall be provided to allow the connection of insulated jumpers to the HV terminals of the circuit breaker. The clamps shall be design to allow the connection of flexible copper insulated jumpers of minimum cross section 70mm².

On the 33kV units insulated jumpers and connectors rated at 800A shall be provided.

3.3.3 Local Mechanical Trip Facility

Facilities shall be provided to allow the circuit breaker to be tripped manually without the need for external power supplies. This shall normally be provided through the use of a yellow pull ring mounted on the exterior of the circuit breaker. The trip ring shall be designed to be operated from ground level with the use of standard insulated operating rods. Once the manual trip has been operated, it shall not be possible to close the circuit breaker either locally or remotely by electrical means without the trip lever being reset.

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3.3.4 Earth Connection

The circuit breaker mounting bracket shall incorporate suitable holes to allow the connection of earth bonding cables.

3.3.5 Erection Facilities

Each unit shall be fitted with adequately rated lifting eyes to facilitate erection onto the pole.

3.4 Anti-Corrosion requirements

Ferrous surfaces shall be protected against corrosion by galvanisation or Zink spray coating. Surfaces shall be prepared in accordance with ENA TS 98-1.

3.4.1 Zink Spray

The zinc shall be applied to a thickness of Zn4 in accordance with BS EN ISO 2063.

3.4.2 Galvanising

Galvanising shall be applied by the hot dip process in accordance with BS EN ISO 1461. Oversize tapping or re-tapping of female threads shall be provided where the bolt or male thread is hot dip galvanised. Nuts shall be tapped up to 0.4mm oversize after galvanising and the threads shall be oiled. Bolts and screwed rods shall be galvanised after thread forming.

3.5 Type and Routine Tests

The equipment shall be subjected to the tests detailed in ENA TS 41-36, ENA TS 48-5 and applicable BS EN and IEC documents detailed in this document. Appendix A (Self Certification Conformance Declaration) details the appropriate clauses and sub-clauses of the relevant standards.

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4. References

4.1 External Documentation

Reference	Title
IEC 60694	Common Specification for HV Switchgear and Control Standard
ENA TS 41-36 Issue 2: 2004	Technical Specification for Distribution Switchgear for Service up to 36kV (cable and overhead connected)
BS EN 62271-100	High-voltage switchgear and controlgear. High-voltage alternating-current circuit-breakers
BS EN 62271-200	High-voltage switchgear and controlgear. AC metal-enclosed switchgear and controlgear for rated voltages above 1 kV and up to and including 52 kV
BS EN 62271-102	High-voltage switchgear and controlgear. High-voltage alternating current disconnectors and earthing switches
IEC 60529	Specification for degrees of protection provided by enclosures (IP code)
ENA TS 48-5	Technical Specification for Protection Approval Panel
ENA TS 98-1	Technical Specification for the Surface Preparation of New Plant
BS EN ISO 2063: 2005	Thermal spraying. Metallic and other inorganic coatings. Zinc, aluminium and their alloys
BS EN ISO 1461: 1999	Hot dip galvanized coatings on fabricated iron and steel articles. Specifications and test methods

4.2 Internal documentation

Reference	Title
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5. Definitions

Term	Definition
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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
S Salkeld	Standards Engineer	S Salkeld	17/01/2006

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G Bolden	Purchasing Manager	G Bolden	17/01/2006

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G Hammel	Overhead Systems Standards Engineer	G Hammel	17/01/2006

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	20/01/2006

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessments Manager	C Holdsworth	17/01/2006

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	17/01/2006

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Appendix A

SELF CERTIFICATION CONFORMANCE DECLARATION

Pole Mounted Auto-reclose Circuit Breakers shall comply with the latest issues of the relevant ENA technical specifications, international and British Standards.

This check sheet identifies the clauses in the aforementioned Standards relevant to Pole Mounted Auto-recloser Circuit Breakers for use on the NEDL and YEDL distribution networks.

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant IEC, 'BS EN' or 'ENATS' as appropriate

Manufacturer:

Product Reference:

Name:

Signature:

Date:

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Appendix A Table

Pole Mounted Auto-Reclosing Circuit Breakers

• Type Testing				
Standard	Clause/Sub-clause	Requirement	Conformance Code	Remarks
IEC 60694	6.2	Dielectric		
IEC 62271-100	6.2			
ENA TS 41-36	Table 1a			
IEC 62271-200	6.2.9	Partial Discharge		
ENA TS 41-18				
IEC 60694	6.4	Measurement of Main Circuit Resistance		
IEC 62271-100	6.4			
IEC 62271-200	6.4			
IEC 60694	6.5	Temperature Rise		
IEC 62271-100	6.5			
IEC 62271-200	6.5			
IEC 60694	6.6	Short-time and Peak Withstand Current		
IEC 62271-100	6.6			
IEC 62271-200	6.6			

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• Type Testing Continued				
Standard	Clause/Sub-clause	Requirement	Conformance Code	Remarks
IEC 60694	6.7	Verification of Protection - Minimum of IP34D Mechanical Impact		
BS EN 662271-100	6.7			
BS EN 62271-200	6.7			
IEC 60529				
ENA TS 41-36	1.5.13			
IEC 60694	6.8	Tightness Test		
BS EN 62271-100	6.8			
BS EN 62271-200	6.8			
IEC 60694	6.9	EMC Test		
BS EN 62271-100	6.9			
BS EN 62271-100	6.101.2.4	Mechanical Operations - class M2 – 10,000 Operating Cycles		
BS EN 62271-200	6.102			
BS EN 62271-100	6.102	Mechanical Strength kinematic chain between moveable contacts and position indicating device		
BS EN 62271-200	6.102			
BS EN 62271-102	6.105			
BS EN 62271-100	6.101.3	Low and High Temperature Tests		
IEC 50056	6.101.5	High Temperature Test – subject to design		
IEC 60129	6.103			
BS EN 62271-100	6.101.5	Operation Under severe Ice conditions		
IEC 6271-102	6.103			

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• Type Testing Continued

Standard	Clause/Sub-clause	Requirement	Conformance Code	Remarks
ANSI /IEEE C37.60	Table 4	Short Circuit Making and Breaking Tests		
BS EN 62271-100	6.111	Cable Charging Current Break Test		
BS EN 62271-100	6.111	Line Charging Current Break tests		
BS EN 62271-200	6.106	Internal Arc		
ENA TS 41-36	1.5.101			
BS EN 62271-200	6.103	Gas Filled Compartment Pressure withstand		
ENA TS 98-1	-	Finish		

• Protection Relay Type Testing – Atmospheric Environment Requirements

Standard	Test	Requirement	Conformance Code	Remarks
IEC 60068-2-1	Temperature Cold Heat	-25°C, 96 hours, operate and storage		
IEC 60068-2-2	Temperature Dry Heat	-70°C, 96 hours, operate and storage		
IEC 60068-2-3	Relative Humidity			
IEC 60068-2-30	Relative Humidity Alternative	100% RH, 40°C, 56 cycles of +25°C to +55°C		
IEC 60529	Enclosure	IP54		

• Protection Relay Type Testing Continued – Mechanical Environment Requirements

Standard	Test	Requirement	Conformance Code	Remarks
IEC 60255-21-1	Vibration	Response and Endurance Class 1		
IEC 60255-21-2	Shock	Response and Endurance Class 1		
IEC 60255-21-2	Bump	Class 1		
IEC 60255-21-3	Seismic	Class 1		

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• Protection Relay Type Testing Continued – Electrical Requirements

Standard	Test	Requirement	Conformance Code	Remarks
IEC 60255-6	48 V DC Supply Voltage	Table 1, 38.5 to 53 V with a maximum of > 60 V		
IEC 60255-11	Voltage Dips, Short Interruptions and Voltage Immunity Test	10ms interruption, 12% AC ripple		
	DC Supply Voltage General	Ramp up and down over 1 minute or similar		
ENA TS 48-4	Low Burden Trip Relays	Captive discharge ENA 1		
ENA TS 48-4	High Burden Trip Relays	Captive discharge ENA 1		
ENA TS 48 – 5	Thermal requirement of CT inputs	2.4 x In, continuous 3.0A, 20 mins 3.5A, 10 mins 4.0A, 5 mins 5.0A, 3 mins 6.0A, 2 mins		
ENA TS 48 – 5	Thermal requirements of VT inputs	120% of Vn, continuous		
ENA TS 48 - 5	Dielectric	Test values selected according to insulation voltage. High Impedance circulating current schemes, test at 2.5kV. Circuits connected to instrument transformers or batteries, rated insulation not below 250V, test at 2.0kV. Open output relay contacts 1kV.		
IEC 60255 - 5	Impulse Voltage	Test at 5kV, 0.5J		

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• Electromagnetic Compatibility (EMC) Requirements					
Standard	Test	Requirement	Conformance Code	Remarks	
IEC 60255	Oscillatory waves immunity test (High Frequency Disturbance)	Class III, 1MHz 2.5kV common, 1 kV diff. Applied to all ports, except diff on comms at the discretion of the approving panel			
IEC 60255	Electrostatic Discharge Immunity Tests	Class III. 6kV, contact, 8kV air. Applied to the enclosure			
IEC 60255	Radiated electromagnetic Field Disturbance test (RFI)	10V/m, 1kHz, 80 to 1000MHz sweep and 80, 160, 450, 900 Mhz spot frequencies. Applied to the enclosure			
IEC 60255	Radiated electromagnetic field from digital radio telephones immunity test	10V/m, 900 and 1890Mhz. Applied to the enclosure			
IEC 60255	Electrical fast transient/burst immunity	Level IV, 4kV. Applied to all ports			
	Surge immunity	Level III, 2kV common, 1kV diff. (Level 4, 4kV, 2kV preferred for CT and VT inputs) Applied to all ports			
IEC 60255	Conducted electromagnetic field disturbance tests	10Vrms, 80% mod, 1kHz. 0.15 to 80MHz sweep and 27 and 68 MHz spot frequencies. Applied to all ports			
EN 61000-4-8 (IEC 61000-4-8)	Power frequency magnetic field immunity test	1000 A/m for 1 sec and 100 A/m for 1 min. Applied to all ports			
EN 61000-4-9 (IEC 61000-4-9)	Pulse magnetic field immunity test	6.4/16 μ s magnetic pulse, 1000 A/m. Applied to enclosure. Not currently mandatory			

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Standard	Test	Requirement	Conformance Code	Remarks
EN 61000-4-10 (IEC 61000-4-10)	Damped oscillatory magnetic field immunity test	0.1 and 1.0 MHz, 100A/m. Applied to the enclosure. Not currently mandatory		
IEC 60255-22-7 & EATS 48-5	Power Frequency	Level 4, 300V for 1 s at 50Hz, common mode. Appendix A, EATS 48-5 for differential, at discretion of panel		
IEC 60834-1 & IEC 60834-2	Communication channel Noise immunity			
IEC 60255-25	Conducted and radiated Emission	Class A, conducted, power supply: 0.15 to 0.5 MHz, 79dB(μV) quasi peak, 66dB (μV) average, 0.5 to 30 MHz, 71dB(μV) quasi peak, 60dB (μV) average. Radiated, enclosure at 10M: 30 to 230 MHz, 40dB(μV) quasi peak, 230 to 1000 MHz, 47dB(μV) quasi peak,		

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Appendix B – Logistical Requirements

1.0 Protection and Packaging

The equipment shall be suitably packaged and protected with each unit marked with the following information:

- (i) Manufacturers name.
- (ii) Manufacturers product type and unique unit serial number.
- (iii) Description of item.
- (iv) Date of supply.
- (v) Company order number.
- (vi) Purchasing Companies Stock Catalogue Number.
- (vii) Weight of each unit.

2.0 Test Certificates

All circuit breakers shall be supplied complete with a copy of the routine tests carried out on the equipment during its manufacture. The test cards shall be enclosed in a waterproof document holder and be tied to the recloser.